Hub: PHY180 Hub
Previous: 5 —2D Kinematics
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Relevant Textbook Sections

4.4 — 4.5

Lecture Notes

Pre-Lecture Notes

Uniform Circular Motion

Uniform Circular Motion deals with the concept of objects rotating around some center with a constant velocity. Note that while the *magnitude* of the velocity vector may remain constant, *its direction* is changing constantly.

Tangential Velocity

Note that with Circular Motion, we can also define the Tangential Velocity of a point as some function if it's angular velocity and the radius of rotation. Recall the definition of Arc Length:

$$a = \theta r$$

Where θ is defined in Radians; then it comes to follow that we can define velocity as some value of $\frac{\Delta\theta}{\Delta t}$, or in other words ω:

$$v_t = \omega r$$

Where ω is in terms of *radians* per second (or some other time unit).

Centripetal Acceleration

Take a circle with a point some distance $\vec{r}$ away from the origin pointing towards some point on the circumference, then:

$$|\vec{r}(t)| = |\vec{r}(t+h)|$$

Note as well, that the velocity of some point at the circumference is also given as:

$$|\vec{v}(t)| = |\vec{v}(t+h)|$$

Then:

$$\Delta\vec{r} = \vec{r}(t+h) - \vec{r}(t)$$

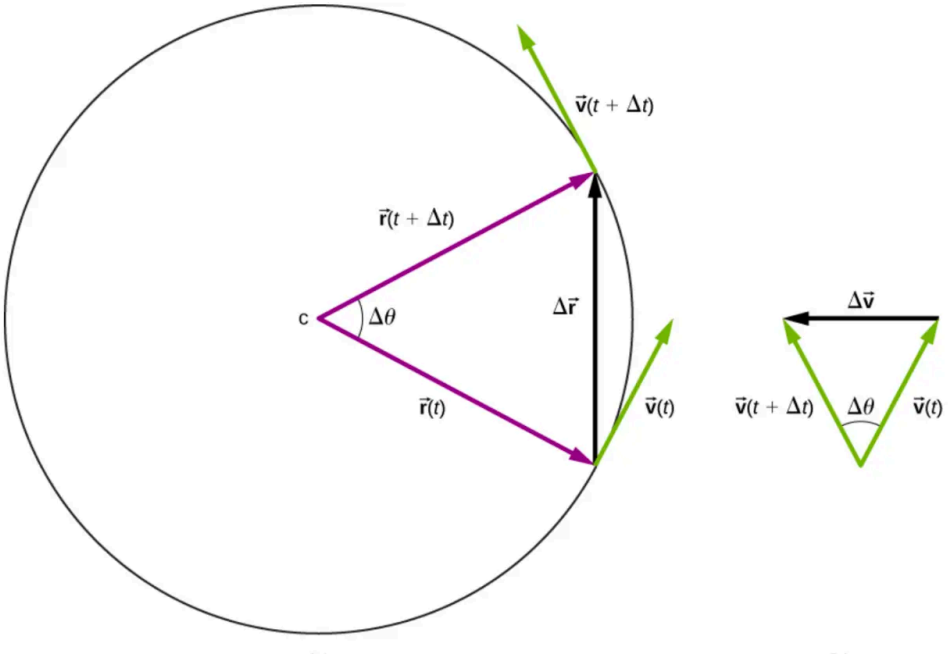
$$\Delta\vec{v} = \vec{v}(t+h) - \vec{v}(t)$$

Based on $\vec{v} \perp \vec{r}$, this implies that the triangle $\vec{v}(t), \vec{v}(t+h), \Delta\vec{v}$ and $\vec{r}(t), \vec{r}(t+h), \Delta\vec{r}$ are similar if $\Delta\theta$ is some really small value, that is the below statements hold true if $\Delta\theta \to 0$, that is:

$$\frac{\Delta v}{\vec{v}} = \frac{\Delta r}{\vec{r}} \implies \Delta v = \frac{\vec{v}\Delta r}{\vec{r}}$$

Or, without vector notation, then

$$\Delta v = \frac{v\Delta r}{r}$$



To solve for a term of *centripetal acceleration*, we can take the derivative of our term:

$$a = \frac{\Delta v}{\Delta t} = \frac{1}{\Delta t}\left(\frac{v\Delta r}{r}\right) = \frac{v}{r}\left(\frac{\Delta r}{\Delta t}\right) = \frac{v^2}{r}$$

That is:

$$a_c = \frac{v^2}{r}$$

Where a_c always points to the center of the circle of rotation, that is it points radially inwards **always**. It can also be termed as a form of **Radial** acceleration.

Equations of Motion

To describe the equations of motion of our particle, we can describe it through $\vec{r}$, and is defined below:

$$\vec{r} = \hat{i}A\sin(\omega t) + \hat{j}A\cos(\omega t)$$

Where ω is angular velocity, in terms of $\frac{rad}{s}$ (or some rotation over time value), and $A = r = |\vec{r}|$.

Note that this is an equation for a term of *displacement*, then we can differentiate it once for velocity, and twice for acceleration respectively. Differentiate with respects to *time*.

$$\vec{v}(t) = \frac{\Delta\vec{r}}{\Delta t} = \hat{i}\omega A\cos(\omega t) - \hat{j}\omega A\cos(\omega t)$$

$$\vec{a}(t) = \frac{\Delta\vec{v}(t)}{\Delta t} = -\hat{i}\omega^2 A\sin(\omega t) - \hat{j}\omega^2 A\cos(\omega t)$$

Note that our earlier definition still holds; $\vec{v}(t)$ is tangent to the circle; and $\vec{a}$ is equivalent to our centripetal acceleration.

Non-Uniform Circular Motion

If our *tangential* velocity is non-constant, then this implies that we have some *acceleration* also tangent to the circle that affects our rotating object, that is:

$$a_t = \frac{d|\vec{v}|}{dt}$$

Note that if $a \neq 0$, then v will change. If a_c is constant; then:

$$a_c = \frac{v^2}{r}$$

For this to remain true, r must scale to balance out the change in $\vec{v}$, meaning that our object's radius will proportionally increase or decrease respectively. Note that if tangential acceleration is present, then the overall acceleration of our point (assuming solely a_c and a_t) is defined by the following:

$$a = a_c + a_t$$

Note that $a_c \perp a_t$; ergo $|\vec{a}| = \sqrt{a_c^2 + a_t^2}$. With $a_t \neq 0$, a will no longer be directly pointing towards the center.

Relative Motion

The rule of thumb for relative motion is to sum up the vector subscripts to get it to a term that results in the reference frame basis that we are looking for: recall that $\vec{AB} = -\vec{BA}$.

Relative Motion can be generally defined by the Galilean Transformation of Velocity:

$$\vec{v}_{CB} = \vec{v}_{CA} + \vec{v}_{AB}$$

i.e. Combine vector subscripts. Note that the Galilean Transformation does not hold under Relativistic speeds, instead the Lorentz Transform is the value which holds.